Hospital Operations & Billing Intelligence Dashboard

Functional Requirements Document(FRD)

1. Project Title

Hospital Operations & Billing Intelligence Dashboard

2. Dashboard Pages / Sections

Page	Purpose	Audience
1. Executive Overview	Headline KPIs and admission and billing trend	Administrator, CFO
2. Billing & Insurance Analysis	Compare billing across insurers, conditions and admission types	CFO, finance team
3. Patient & Operations Insights	Length of stay, age groups, test results, admission mix	Operations manager

3. Data Requirements

- Table: Admissions_Clean (50,000 rows, 18 columns). One row = one patient admission.
- Columns used: Admission ID, Age, Age Group, Gender, Medical Condition, Date of Admission, Discharge Date, Insurance Provider, Billing Amount, Billing Status, Admission Type, Medication, Test Results, Length of Stay(Day)
- Columns not used for ranking: Doctor, Hospital (about one value per row)
- A separate Date table will be added for time intelligence.

4. Filters / Slicers

Slicer	Field	Pages	Behaviour
Year	Date table [Year]	All (synced)	Filters every visual
Medical Condition	Medical Condition	All (synced)	Multi-select
Insurance Provider	Insurance Provider	All (synced)	Multi-select
Admission Type	Admission Type	All (synced)	Multi-select
Age Group	Age Group	Page 3	Multi-select

5. Visuals / Charts

Page	Visual	Fields	Why
1	6 KPI cards	The 6 KPI measures	Instant headline numbers
1	Line chart	Month (axis), Total Admissions and Total Billing	Trend over time

1	Donut	Admission Type, Total Admissions	Admission mix
2	Clustered bar	Insurance Provider, Total Billing	Compare insurers
2	Bar	Medical Condition, Avg Billing per Admission	Cost per condition
2	Matrix	Insurance Provider × Admission Type, Avg Billing	Spot gaps
2	Table/card	Billing Status = Negative: count and amount	Refund and adjustment tracking
3	Column chart	Age Group, Avg Length of Stay	Stay by age
3	Bar	Medical Condition, Avg Length of Stay	Stay by condition
3	Donut	Test Results, Total Admissions	Severity mix

6. Interactivity

- Cross-filtering between all visuals on a page
- Synced slicers across pages
- Drill-down on the trend chart: Year → Quarter → Month
- Tooltips: a report-page tooltip showing admissions, billing and average stay
- Page navigation buttons (bookmarks) and a "reset filters" button
- Drill-through from Medical Condition to a detail table of admissions

7. Calculations / Measures (table: Admissions_Clean; expected values on all data)

Measure	**DAX logic**	**Expected**	**Measure**
Total Admissions	COUNTROWS(Admissions_Clean)	50,000	Total Admissions
Total Billing	SUM(Admissions_Clean[Billing Amount])	1,277,784,576.67	Total Billing
Avg Billing per Admission	DIVIDE([Total Billing],[Total Admissions])	25,555.69	Avg Billing per Admission
Avg Length of Stay	AVERAGE(Admissions_Clean[Length of Stay(Day)])	15.5 days	Avg Length of Stay
Emergency Admission %	DIVIDE(CALCULATE([Total Admissions], Admissions_Clean[Admission Type]="Emergency"), [Total Admissions])	32.78%	Emergency Admission %

Abnormal Test Result %	same pattern with Test Results = “Abnormal”	33.54%	Abnormal Test Result %
Negative Billing Count	CALCULATE([Total Admissions], Admissions_Clean[Billing Status]<>"Normal")	96	Negative Billing Count

Time-intelligence measures (Billing YTD, same-period-last-year) are added with the Date table on Day 5.

8. Export / Reporting Needs

PDF export of all 3 pages saved as PowerBI/exports/dashboard_export.pdf.

9. Acceptance Criteria

1. KPI cards match the expected values above when no slicer is applied.
2. Every slicer filters all visuals on all pages (except where a page-only slicer is listed).
3. Totals by insurer add up to Total Billing, and admissions by admission type add up to 50,000.
4. Drill-down, drill-through, tooltips and navigation buttons work.
5. Page 1 numbers reconcile to the cleaned CSV (50,000 rows).
6. Visuals use a consistent theme, titles and readable labels.

10. Notes / Special Instructions

- Data appears synthetic; insights are for demonstration.
- 2019 and 2024 are partial years: avoid year-on-year claims for them.
- Currency is not stated in the data; amounts are shown without a currency symbol.
- Patient names are removed for privacy.